Practice 6 — Combinations and Binomial Coefficients

Solutions to Selected Problems

Problems 8, 9, 11

Discrete Mathematics — QYSU

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1 Problem 8 — Lattice Paths Through a Given Point

Theory Recap: Lattice Paths and Combinations

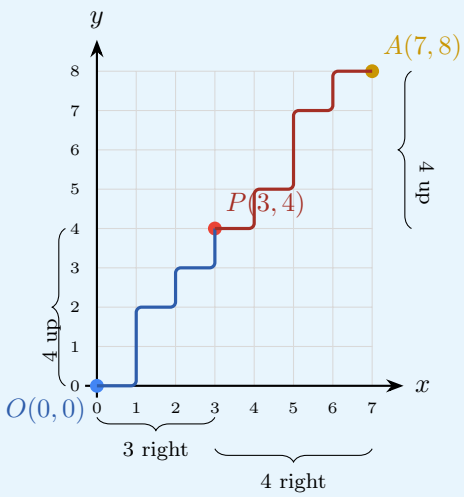
A **lattice path** from $(0,0)$ to (m,n) consists of exactly m steps right (R) and n steps up (U), in some order. The total number of steps is $m+n$, and we must choose which m of them are R-steps (the rest are U-steps):

$$\text{Number of paths from } (0,0) \text{ to } (m,n) = \binom{m+n}{m} = \binom{m+n}{n}.$$

Paths through an intermediate point. If a path must pass through a point $P(p,q)$ on the way from $O(0,0)$ to $A(m,n)$ (with $0 \le p \le m$ and $0 \le q \le n$), we decompose the journey into two independent segments:

$$O \to P: \binom{p+q}{p} \text{ paths}, \qquad P \to A: \binom{(m-p)+(n-q)}{m-p} \text{ paths}.$$

By the **multiplication principle**, the total count is the product.



Problem Statement

Find the number of lattice paths from $O(0,0)$ to $A(7,8)$ that pass through the point $P(3,4)$.

Solution

We split the path into two independent segments at $P(3,4)$.

Segment 1: $O(0,0) \to P(3,4)$. We need 3 steps right and 4 steps up, for a total of 7 steps. Choose which 3 are right-steps:

$$\binom{3+4}{3} = \binom{7}{3} = \frac{7!}{3! \cdot 4!} = \frac{7 \cdot 6 \cdot 5}{3 \cdot 2 \cdot 1} = 35.$$

Segment 2: $P(3,4) \to A(7,8)$. From $(3,4)$ to $(7,8)$ we need $7-3=4$ steps right and

$8 - 4 = 4$ steps up, for a total of 8 steps. Choose which 4 are right-steps:

$$\binom{4+4}{4} = \binom{8}{4} = \frac{8!}{4! \cdot 4!} = \frac{8 \cdot 7 \cdot 6 \cdot 5}{4 \cdot 3 \cdot 2 \cdot 1} = 70.$$

Multiplication principle. The two segments are independent (any path for the first segment can be combined with any path for the second), so:

$$\text{Total paths} = 35 \times 70 = 2450.$$

The number of lattice paths from $O(0,0)$ to $A(7,8)$ passing through $(3,4)$ is $\binom{7}{3} \cdot \binom{8}{4} = 35 \cdot 70 = \boxed{2450}$.

2 Problem 9 — Partitioning 6 into Three Summands

Theory Recap: Stars and Bars / Integer Compositions and Partitions

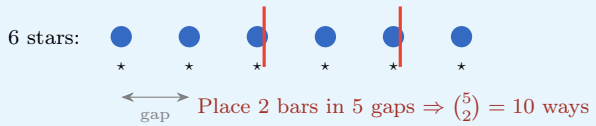
Stars and bars (combinations with repetition). The number of solutions to

$$x_1 + x_2 + \cdots + x_k = n, \qquad x_i \geq 0,$$

in non-negative integers is $\binom{n+k-1}{k-1}$.

If we require $x_i \geq 1$ (positive integers), substitute $y_i = x_i - 1 \geq 0$ to get $y_1 + \cdots + y_k = n - k$, giving $\binom{n-1}{k-1}$ solutions.

Example: $x_1 + x_2 + x_3 = 6$, $x_i \geq 1$



Integer partitions (unordered). A partition of n into exactly k parts is a way to write $n = \lambda_1 + \lambda_2 + \cdots + \lambda_k$ with $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k \geq 1$. Order does not matter, so $4 + 1 + 1$ and $1 + 4 + 1$ are the *same* partition.

Problem Statement

Find the number of ways to partition the number 6 into three summands.

Solution

This problem admits two natural interpretations depending on whether the order of summands matters.

Interpretation 1: Ordered summands (compositions)

We seek the number of solutions to

$$x_1 + x_2 + x_3 = 6, \qquad x_i \geq 1.$$

By the stars-and-bars formula with the substitution $y_i = x_i - 1$:

$$\binom{6-1}{3-1} = \binom{5}{2} = \frac{5!}{2! \cdot 3!} = 10.$$

#	x_1	x_2	x_3
1	4	1	1
2	1	4	1
3	1	1	4
4	3	2	1
5	3	1	2
6	2	3	1
7	1	3	2
8	2	1	3
9	1	2	3
10	2	2	2

Interpretation 2: Unordered summands (integer partitions)

We seek the number of ways to write $6 = \lambda_1 + \lambda_2 + \lambda_3$ with $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq 1$.

Enumerating all possibilities:

#	**Partition**
1	$4+1+1$
2	$3+2+1$
3	$2+2+2$

There are exactly **3** such partitions. We verify completeness: $\lambda_1 \geq 2$ always (since $\lambda_1 \geq \lceil 6/3 \rceil = 2$). If $\lambda_1 = 4$: $\lambda_2 + \lambda_3 = 2$, so $\lambda_2 = \lambda_3 = 1$. If $\lambda_1 = 3$: $\lambda_2 + \lambda_3 = 3$ with $3 \geq \lambda_2 \geq \lambda_3 \geq 1$, giving $\lambda_2 = 2, \lambda_3 = 1$. If $\lambda_1 = 2$: $\lambda_2 + \lambda_3 = 4$ with $2 \geq \lambda_2 \geq \lambda_3 \geq 1$, giving $\lambda_2 = 2, \lambda_3 = 2$.

Ordered (compositions with positive parts): $\binom{5}{2} = 10$ ways.

Unordered (integer partitions into 3 parts): 3 ways: 4+1+1, 3+2+1, 2+2+2.

3 Problem 11 — Rearrangements of "matematika"

Theory Recap: Multinomial Coefficient / Permutations with Repetition

If we have n objects where there are k distinct types with multiplicities $n_1, n_2, \ldots, n_k$ (so $n_1 + n_2 + \cdots + n_k = n$), the number of distinct arrangements is the **multinomial coefficient**:

$$\binom{n}{n_1,\ n_2,\ \ldots,\ n_k} = \frac{n!}{n_1! \cdot n_2! \cdots n_k!}.$$

Intuition. Start with $n!$ arrangements of all objects treated as distinguishable. Then divide out the $n_i!$ internal permutations for each group of identical objects.

$$\boxed{n!\text{ total arrangements}} \quad \div \quad \boxed{n_1!} \quad \boxed{n_2!} \quad \cdots \quad \boxed{n_k!} \quad = \quad \boxed{\text{distinct arrangements}}$$

Problem Statement

How many different words can be formed by rearranging all the letters of the word **"matematika"**?

Solution

Step 1: Identify the letters and their frequencies.

The word "matematika" has the letters:

$$\text{m} - \text{a} - \text{t} - \text{e} - \text{m} - \text{a} - \text{t} - \text{i} - \text{k} - \text{a}$$

This gives $n = 10$ letters in total. Counting each:

Letters:	**m**	**a**	**t**	**e**	**m**	**a**	**t**	**i**	**k**	**a**
Position:	1	2	3	4	5	6	7	8	9	10

Letter	a	m	t	e	i	k
Count n_i	3	2	2	1	1	1

Verification: $3 + 2 + 2 + 1 + 1 + 1 = 10$ ✓

Step 2: Apply the multinomial coefficient.

$$\frac{10!}{3! \cdot 2! \cdot 2! \cdot 1! \cdot 1! \cdot 1!} = \frac{10!}{3! \cdot 2! \cdot 2!}$$

Computing:

$$\begin{aligned} 10! &= 3\,628\,800, \\ 3! \cdot 2! \cdot 2! &= 6 \cdot 2 \cdot 2 = 24, \\ \frac{3\,628\,800}{24} &= 151\,200. \end{aligned}$$

The number of distinct rearrangements of "matematika" is $\dfrac{10!}{3! \cdot 2! \cdot 2!} = \boxed{151\,200}$.